

Impact Evaluation: Test

75 minutes, February 2026

- Read the instructions and questions carefully
- Please be mindful of CEU policies on plagiarism and academic dishonesty

Long questions (30 points each), 60 points

1. The Governor of California is considering policy changes based on the relationship between class size and student test scores. His team conducted a regression analysis using observational data from California school districts and reported their initial findings (Column 1). Upon controlling for the number of students enrolled in the free lunch program, the negative relationship between class size and test scores appeared less pronounced (Column 2). This change prompted the Governor to seek a second opinion. An attentive staff member suggested comparing these results with data from Project STAR, a randomized controlled trial (RCT) from Tennessee (Columns 3 and 4). Before enacting any new policies on class size, the Governor has enlisted the help of a CEU causal inference student to provide additional insights.

Using the regression results provided in the table below, please advise the Governor:

Table 1: Observational vs. Experimental Data

	<i>Dependent variable: Average student scores</i>			
	CA Schools		STAR Project	
	(1)	(2)	(3)	(4)
Large class	−2.280*** (0.480)	−1.117*** (0.240)	−6.559*** (2.036)	−7.040*** (1.960)
Free lunch		−0.600*** (0.017)		−40.002*** (1.866)
Observations	420	420	5,786	5,786

Note: Treatment variable is the large class vs. small class in STAR.

In CA Schools the treatment variable is continuous: student-teacher ratio.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

- (a) Can we infer causal effects of class size on test scores from the California schools' observational data? Why are the coefficients changing drastically from col. 1 to 2?

- (b) How do the results from the STAR project inform us about causal effects? Why no big change from col. 3 to 4?
- (c) What policy recommendations would you make based on these findings?
- (d) Draw a causal graph depicting how randomization of class sizes (as in the STAR experiment) could address issues arising from observational data – use the outcome variable (test scores), identifying the endogenous variable (X) and omitted variable (O).
2. On April 1, 1992, New Jersey implemented a law raising the minimum wage from \$4.25 to \$5.05 per hour. During the same period, the minimum wage in neighboring Pennsylvania did not change. Card and Krueger (1994) analyzed employment data from fast-food restaurants in both states, collected before and after New Jersey's wage increase. In this scenario, the 'treatment' refers to the minimum wage hike; New Jersey's fast-food industry represents the 'treated' group, whereas Pennsylvania's serves as the 'control' group. The regression results presented focus on the number of full-time-equivalent (FTE) employees as the outcome variable, both before and after the law's enactment.

	before	after
(Intercept)	23.33 *** (1.10)	21.17 *** (1.04)
nj	-2.89 * (1.23)	-0.14 (1.16)
R ²	0.01	0.00
Adj. R ²	0.01	-0.00
Num. obs.	398	396

*** p < 0.001; ** p < 0.01; * p < 0.05.
 nj - dummy variable if it's New Jersey state.

- (a) Why not just compare post-treatment outcomes in New Jersey to Pennsylvania to estimate the effect of minimum-wage law (reported in col. 2, under *after*)?
- (b) Estimate the difference in differences using the results from the two models. What is your conclusion on the effect of minimum wage laws on (fte) employment?
- (c) How would you examine the validity of parallel trend assumption?
- (d) If there is evidence of non-parallel pre-treatment trends between New Jersey and Pennsylvania, how might this affect the DID estimate, and what alternative strategies could be used to assess the causal effect of the minimum wage increase?

Answer only 3 of 4 (10 points each, max 50 words per answer), 30 points

1. Here is a causal graph from nutrition experts: $JunkFood(J) \rightarrow Cholesterol(C) \rightarrow HeartDisease(H)$. A data analyst provides the following regression outputs with heart disease as the dependent variable. Which of the two models provides a less biased estimate of the effect of junk food on heart disease and why?

	Model 1	Model 2
(Intercept)	16.72*** (3.83)	6.27*** (1.57)
J	5.78*** (0.25)	-0.18 (0.27)
C		3.04*** (0.13)
R ²	0.84	0.98
Adj. R ²	0.84	0.98
Num. obs.	100	100

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

2. What key assumption allows us to estimate causal effects with matching? Explain briefly using z (observed), u (unobserved), treatment (D), and y (outcome) in your response.
3. What key assumption underpins the ability to estimate causal effects using Two-Way Fixed Effects (TWFE) models in panel data (e.g., when estimating the impact of beer taxes on traffic accidents across the US states)?
4. Explain the possible tradeoffs when drawing causal inference from data resulting from exact matching vs. coarsened exact matching (hint: variance vs. bias).

Please type the following sentence at the top of your exam, followed by your Student ID Number.

"I certify this is my work, and I have neither given nor received help from anyone else while writing this exam"

- Student ID Number: